

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application
Practical – X

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Class :- TYIT	Batch :-
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PART B – Questions

A) Create a PWA with a Web App Manifest file including: App name Icons Start URL Display mode Implement Push Notifications in a PWA.

Code:-

Program.cs:-

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();
app.UseDefaultFiles();
app.UseStaticFiles();
app.Run();
```

manifest.json:-

```
{
  "name": "E-Commerce PWA",
  "short_name": "ShopPWA",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

Index.html:-

```
<!DOCTYPE html>
<html>
<head>
  <title>E-Commerce PWA</title>

  <link rel="manifest" href="manifest.json">

  <meta name="theme-color" content="#0466d6">
</style>
  body {
    font-family: Arial;
    margin: 20px;
  }

  .container {
```

```
    display: flex;
    gap: 30px;
  }

.cart-section {
  width: 30%;
  border: 1px solid gray;
  padding: 15px;
  min-height: 300px;
}

.product-section {
  width: 70%;
}

.products {
  display: flex;
  flex-wrap: wrap;
  gap: 15px;
}

.product {
  border: 1px solid gray;
  padding: 10px;
  width: 200px;
}

button {
  padding: 5px 10px;
  cursor: pointer;
}
</style>
</head>
<body>
  <h1>E-Commerce PWA</h1>
  <div class="container">

    <!-- Cart Section -->
    <div class="cart-section">
      <h2>Cart Items</h2>
      <ul id="cartList"></ul>
    </div>

    <!-- Product Section -->
    <div class="product-section">

      <h2>Products</h2>

      <div class="product">
        <h3>Laptop</h3>
        <p>Price: $ 5000</p>
        <button onclick="addToCart('Laptop')">
          Add to Cart
        </button>
      </div>

      <div class="product">
```

```

    <h3>Mobile</h3>
    <p>Price: $ 2000</p>
    <button onclick="addToCart('Mobile')">
      Add to Cart
    </button>
  </div>

```

```

<div class="product">
  <h3>Headphones</h3>
  <p>Price: $ 3000</p>
  <button onclick="addToCart('Headphones')">
    Add to Cart
  </button>
</div>

```

```

</div>

```

```

<script src="app.js"></script>

```

```

</body>

```

```

</html>

```

App/js:-

```

if ('serviceWorker' in navigator) {
  navigator.serviceWorker.register('service-worker.js')
    .then(() => console.log('Service Worker Registered'));
}

```

```

let cart = JSON.parse(localStorage.getItem('cart')) || [];

```

```

displayCart();

```

```

function addToCart(product) {
  cart.push(product);

  localStorage.setItem('cart', JSON.stringify(cart));

  displayCart();
}

```

```

function displayCart() {
  let cartList = document.getElementById('cartList');

  cartList.innerHTML = "";

  cart.forEach(item => {
    let li = document.createElement('li');

    li.textContent = item;

    cartList.appendChild(li);
  });
}

```

Service-worker.js:-

```

const CACHE_NAME = 'ecommerce-cache-v1';

```

```

const urlsToCache = [
  '/',
  '/index.html',

```

```

'/app.js',
'/manifest.json'
];

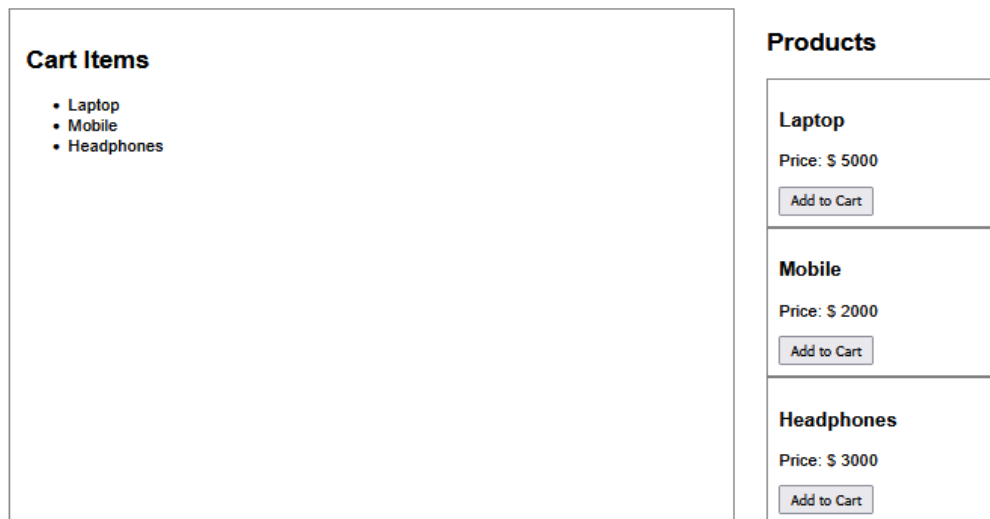
self.addEventListener('install', event => {
  event.waitUntil(
    caches.open(CACHE_NAME)
      .then(cache => {
        return cache.addAll(urlsToCache);
      })
  );
});

self.addEventListener('fetch', event => {
  event.respondWith(
    caches.match(event.request)
      .then(response => {
        return response || fetch(event.request);
      })
  );
});

```

Output:-

E-Commerce PWA



B) Develop an E-Commerce PWA with product listing and offline cart functionality.

Output:-

